FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2019

TEST DATE: 16-12-2018

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. At higher temperatures, specific heat is also higher. Therefore, block with higher initial temperature will suffer less change in temperature.

2. B

Sol. $\lambda = \dfrac{h}{mv} \Rightarrow \lambda \propto 1/v$

$v = \sqrt{v_y^2 + v_0^2}$

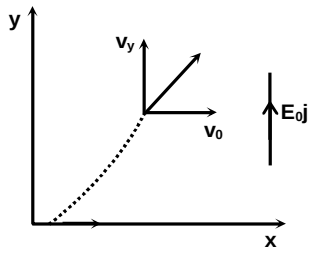
$v_y = u_y + a_y t$

$v_y = 0 + \dfrac{qE_0}{m} t$

$(3v_0) = \sqrt{v_y^2 + v_0^2}$

$\Rightarrow v_y^2 = 8v_0^2 \quad \Rightarrow \dfrac{qE_0}{m} t = 2\sqrt{2} v_0$

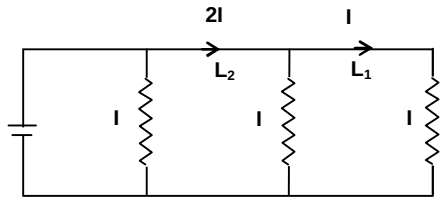
$t = \dfrac{2\sqrt{2}m}{qE_0} v_0 \Rightarrow t \propto 1/E_0$



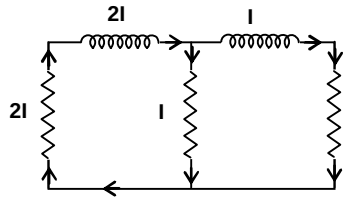
3. A

Sol. When switch is closed

Initially circuit is in steady state current through each resistor as all are identical and are in parallel combination. When switch is off current through L_1 and L_2 just after unchanged. So flow of current remain same.



In right and middle wire current is I downward and in left wire current is 2I upward



4. A

Sol. $I_{cm} + m(R + y)^2 = I_3$ …(i)

$I_{cm} + m(R - y)^2 = I_1$ …(ii)

from (i) and (ii)

$I_1 - I_3 = m[(R - y)^2 - (R + y)^2]$

$I_1 - I_3 = m(2R)(-2y)$ …(iii)

$I'_{cm} + m(R + x)^2 = I_4$ …(iv)

$I'_{cm} + m(R - x)^2 = I_2$ …(v)

From (iv) and (v)

$(I_2 - I_4) = m[(R - x)^2 - (R + x)^2]$

$(I_2 - I_4) = m[(2R)(-2x)]$ …(vi)

from (iii) and (vi)

distance from CM from O

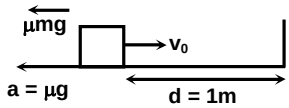
$= \sqrt{x^2 + y^2}$

$= \frac{1}{4MR}\sqrt{(I_1 - I_3)^2 + (I_2 - I_4)^2}$, putting values d = 5 m

5. A

Sol. Velocity after time t, $v = v_0 - \mu gt$

$\Rightarrow v_0 > 2\mu g$ …(i)



Displacement in time 't'

$S = v_0 t - \frac{1}{2}\mu g t^2$

$1 = 2v_0 - \frac{1}{2}\mu g(2)^2 \quad \Rightarrow v_0 = \left(\frac{1}{2} + \mu g\right)$ …(ii)

From (i) and (ii)

$\frac{1}{2} + \mu g > 2\mu g$

$\Rightarrow \quad \mu < 1/2g$

$\Rightarrow \quad \mu < 0.05$

6. A

Sol. $45 + r > \theta_C$ …(i)

$45 - r > \theta_C$ …(ii)

$90° > 2\theta_C$

$45° > \theta_C$

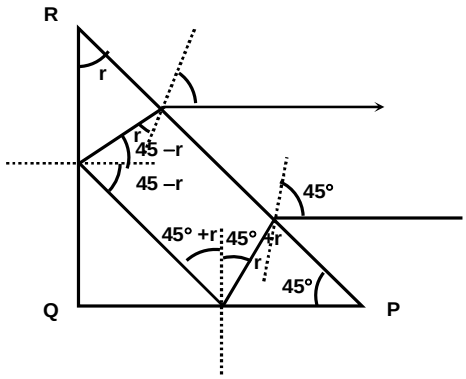
$\sin 45° > \sin \theta_C$

$\frac{1}{\sqrt{2}} > \frac{1}{\mu}$

$\mu > \sqrt{2}$

taking equation 2 only

$45 - \theta_C > r$, $\sin(45 - \theta_C) > \sin r$



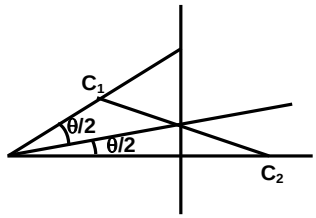
$$\frac{1}{\sqrt{2}}\cos\theta_C - \frac{1}{\sqrt{2}}\sin\theta_C > \frac{\sin 45°}{\mu}$$

$$\frac{\sqrt{\mu^2-1}}{\mu} - \frac{1}{\mu} > \frac{1}{\mu} \Rightarrow \sqrt{\mu^2-1} > 2 \Rightarrow \mu > \sqrt{5}$$

7. A

Sol. $\ell\cos\theta = \frac{\ell}{2}\cos^2\frac{\theta}{2}$

$\cos\theta = 1/3$



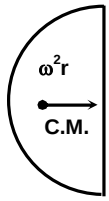
8. D

Sol. $v = \sqrt{\frac{T}{\mu}}$ and $A_t = \left(\frac{2v_2}{v_2+v_1}\right)A_i$ and $A_r = \left(\frac{v_2-v_1}{v_2+v_1}\right)A_i$

9. C

Sol. r = 3R/8

$$F = \frac{m}{2}\cdot\left(\frac{v}{R}\right)^2\frac{3R}{8} = \frac{3mv^2}{16R} = 3N$$



10. C

Sol. Associative Law

$(W + X)\cdot(W + Y) = W + (X\cdot Y)$

11. B

Sol. Use, $N = \frac{\rho v d}{\eta}$

12. C

Sol. In steady state,

$\sigma AT^4 = 2\,\sigma A(300)^4$

and $P + \sigma A(300)^4 = 2\,\sigma AT^4$

$\therefore P = 3\sigma A(300)^4 = 3$ kW

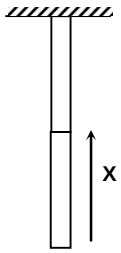
13. D

Sol. Wave speed $v = \sqrt{gx}$

$$\frac{dv}{dt} = \sqrt{g}\cdot\frac{1}{2\sqrt{x}}\cdot\sqrt{gx} = \frac{g}{2}$$

$$\therefore\ A(2\pi f)^2 = \frac{g}{2}$$

i.e., $8\pi^2 f^2 A = g$



14. C

Sol. $|M| = \frac{v_0}{u_0}\left(1+\frac{D}{f_e}\right) = \frac{11}{2}$

$\therefore\ S_{min} = \frac{0.22}{11/2} = 0.04$ mm

15. A

Sol. $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{\sigma}{\rho}} = \frac{1}{2\ell}\sqrt{\frac{Y\varepsilon}{\rho}}$

16. C

Sol. $P = Ei - i^2r$

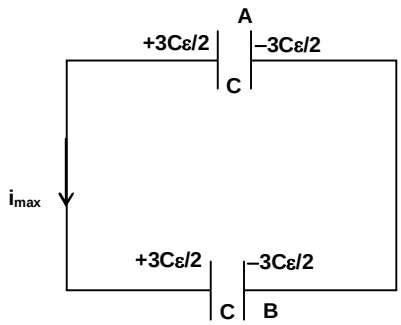
and $i = \frac{E}{2r}$ for P_{max}

17. D

Sol. $\frac{P_n}{P_0} \propto C^n$, n : no of strokes

18. B

Sol. Using conservation of energy



$\frac{1}{2}C(2\varepsilon)^2 + \frac{1}{2}C\varepsilon^2 = \frac{1}{2}(C+C)\left(\frac{3\varepsilon}{2}\right)^2 + \frac{1}{2}Li_{max}^2$

$2C\varepsilon^2 + \frac{C\varepsilon^2}{2} = \frac{9C\varepsilon^2}{4} + \frac{1}{2}Li_{max}^2$

$\frac{5C\varepsilon^2}{2} - \frac{9C\varepsilon^2}{4} = \frac{1}{2}Li_{max}^2$

$\frac{C\varepsilon^2}{4} = \frac{1}{2}Li_{max}^2$

$i_{max} = \varepsilon\sqrt{\frac{C}{2L}}$

19. A

Sol. In equilibrium $(m + m') = \frac{1}{3}\pi\left(\frac{h}{2}\right)^3 \tan^2\alpha\rho g$ (α : semi-vertical angle of the cone)

When displaced slightly by y

$f_r = (m+m')g - \frac{1}{3}\pi\left(\frac{h}{2}+y\right)^3 \tan^2\alpha\rho g$

$\therefore\ a = -\frac{6gy}{h}$

$\therefore\ f = \frac{1}{2\pi}\sqrt{\frac{6g}{h}}$

20. C

Sol. $(m_1 + m_2)g = KL$

$\frac{1}{2}KL^2 = m_1g \cdot 2L$

21. D

Sol. Consider a pyramid of square base of side $\sqrt{2}a$.

22. B

Sol. $\int L_1 di_1 i_1 + \int L_2 di_2 i_2 + \int (Mdi_1 i_2 + Mdi_2 i_1)$

$= \int L_1 di_1 i_1 + \int L_2 di_2 i_2 + \int Md(i_1 i_2) = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + Mi_1 i_2$

23. B

Sol. M.S. reading 35 mm

X + 4V = nM

$X = nM - 4\left(\frac{5}{8}M\right)$

X < 1 mm for n = 3 only

So, $X = 3 - \frac{5}{2} = \frac{1}{2}$mm

$\therefore$ reading = 35 + 0.5 mm

= 3.55 cm

24. B

Sol. Maximum number of β particles will be emitted when dn/dE = 0

i.e. at E = $\frac{E_{max}}{5}$

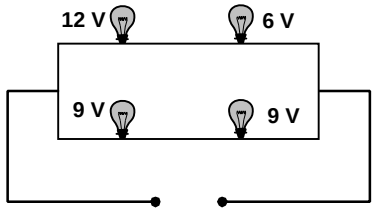
Thus, maximum number of neutrinos would come out energy $\frac{4}{5}E_{max}$.

25. D

Sol. Maximum p.d. can be 18 V

$R_{eq} = \frac{9\times 4}{9+4} = \frac{36}{13}\Omega$

$\therefore P_{max}$ = 117 W



26. A

Sol. In equilibrium T = mg and RT = $\pi R^2 BI$

27. A

Sol. For central maxima

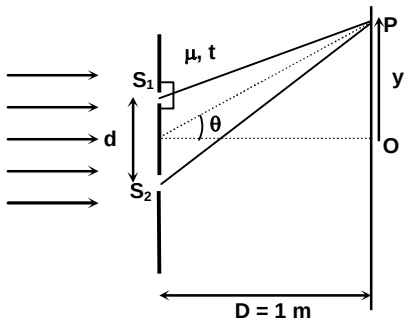
$(\mu - 1)t = d\sin\theta$

$\sin\theta = \frac{(\mu-1)t}{d} = \frac{(1.5-1)1\times 10^{-5}}{1\times 10^{-3}} = \frac{1}{200}$

$\Rightarrow$ for small θ

$\tan\theta = \frac{1}{200}$, $\frac{y}{D} = \frac{1}{200}$

$\therefore$ y = 0.5 cm



28. D

Sol. Modulation index $\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$

29. B

Sol. Basic Concept

30. A

Sol. $T = \frac{PV}{R}$

T will be maximum when PV is maximum

$T = \frac{PV}{R} = \frac{(4 + 2\sin\theta)(4 + 2\cos\theta)}{R}$

As sin θ and cosθ both can not be equal to 1 for same value of θ

$\therefore$ T can not be 36/R

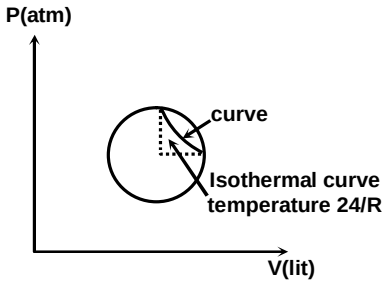
T_{max} should be less than 36R

Curve is above isothermal curve

$\therefore$ temperature is more than 24/R on the given process

So T_{max} lie between 24/R and 36/R

Only one option is present.



Chemistry

PART – II

SECTION – A

31. B

Sol. In ccp number of O^{2-} ions per unit cell = 4

Total negative charge per unit cell = 4 × 2 = 8

Number of tetrahedral voids = 4 × 2 = 8

Number of octahedral voids = 4 × 1 = 4

Number of tetrahedral voids occupied $= 8 \times \frac{12.5}{100} = 1$

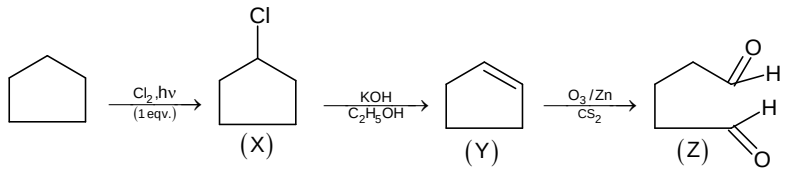
Number of negative charges left unneutralized = 8 – 2 = 6

Number of Y^{3+} ions required to occupy octahedral voids $= \frac{6}{3} = 2$

Fraction of octahedral voids occupied $= \frac{2}{4} = 0.5$

32. D

Sol.



33. B

Sol. (A) Li when burnt in air only forms monoxide.

(B) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

(C) Lithium has high I.E. as compared to other alkali metals and imparts crimson red colour to flame.

(D) Li_2SO_4 is the only alkali metal sulphate which does not form double salts.

34. A

Sol. (A) $XeF_4 + O_2F_2 \longrightarrow XeF_6 + O_2$

(B) $XeF_4 + H_2 \longrightarrow Xe + HF$

(C) $XeF_4 + KI \longrightarrow KF + Xe + I_2$

(D) $XeF_4 + SF_4 \longrightarrow SF_6 + Xe$

35. B

Sol. (A) $\left[Al(OH)_4\right]^-(aq) + NH_4^+(aq) \longrightarrow \underset{(White)}{Al(OH)_3}\downarrow + NH_3\uparrow + H_2O$

(B) $Pb^{2+}(aq) + 2Br^-(aq) \longrightarrow PbBr_2\downarrow(White)$

(C) $BiI_3\downarrow(\text{black precipitate}) + H_2O \xrightarrow{\Delta} BiOI(\text{orange})$

(D) $S^{2-} + Na_2\left[Fe(CN)_5NO\right] \longrightarrow \underset{(\text{Purple colouration})}{Na_4\left[Fe(CN)_5NOS\right]}$

36. B

Sol. $InOH \rightleftharpoons In^+ + OH^-$

$K_b = \frac{[In^+][OH^-]}{[InOH]}$

Also $pOH = pK_b + \log\frac{[In^+]}{[InOH]}$

At pH = 3.3, pOH = 14 – 3.3 = 10.7, the indicator will be sufficiently ionized to give the colour solely due to In^+ ion.
Hence,

$10.7 = pK_b + \log\frac{[In^+]}{[InOH]}$

$= (14 - pk_a) + \log\frac{[In^+]}{[InOH]}$

$= \left[14 - \frac{(3.3 + 4.7)}{2}\right] + \log\frac{[In^+]}{[InOH]}$

$\Rightarrow \log\frac{[In^+]}{[InOH]} = 0.7 = \log 5$

$\Rightarrow \frac{[In^+]}{[InOH]} = 5$

37. B

Sol. Amount of $H_2C_2O_4$ in 200 cm^3 of original solution $= \frac{200}{1000} \times 0.5 \times 90 = 9.0$ g

Molarity of filterate $= \frac{32 \times 0.25}{2 \times 10} = 0.4$ M

Amount of the acid left out after adsorption = 0.4 × 0.2 × 90 = 7.2 g
Amount of the acid adsorbed = 9.0 – 7.2 = 1.8 g

Amount of the acid adsorbed (x/m) per gram $= \frac{1.8}{9} = 0.2$

38. A

Sol. $E^o_{Fe^{3+}/Fe^{2+}} = 3E^o_{Fe^{3+}/Fe} - 2E^o_{Fe^{2+}/Fe}$

= 3 × (- 0.04) – 2 × (- 0.44) V
= 0.76 V

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$

$E_{Fe^{3+}/Fe^{2+}} = E^o_{Fe^{3+}/Fe^{2+}} - \frac{2.303RT}{F}\log\frac{[Fe^{2+}]}{[Fe^{3+}]}$

$0.718 = 0.76 - 0.06\log\frac{[Fe^{2+}]}{[Fe^{3+}]}$

$\Rightarrow \log\frac{[Fe^{2+}]}{[Fe^{3+}]} = \frac{0.042}{0.06} = 0.7 = \log 5 \quad \Rightarrow \frac{[Fe^{2+}]}{[Fe^{3+}]} = 5$

39. A

Sol. Meq. of H_2SO_4 taken to absorb $NH_3 = 50 \times 0.2 = 10$

Meq. of H_2SO_4 left after NH_3 absorption = Meq. of NaOH used for $H_2SO_4 = 25 \times 0.19 = 4.75$

$\therefore$ Meq. of H_2SO_4 used for $NH_3 = 10 - 4.75 = 5.25$

$$\therefore \%N = \frac{1.4 \times NV}{\text{wt. of Fertilizer}} = \frac{1.4 \times 5.25}{1.731} = 4.2\%$$

40. C

Sol. Meq. of H_2SO_4 (original) $= 30 \times 1 = 30$ meq.

Meq. of H_2SO_4 after passing $NH_3 = 30 \times 0.2 = 6$

Meq. of H_2SO_4 reacted = Meq. of $NH_3 = 30 - 6 = 24$

$$\therefore \frac{W_{NH_3}}{17} \times 1000 = 24 \Rightarrow W_{NH_3} = 0.408 \text{ g}$$

$$\therefore \text{Volume of } NH_3 \text{ at STP} = \frac{0.408}{17} \times 22.4 = 0.5376 \text{ lit}$$

= 537.6 ml.

41. D

Sol. Diborane forms adduct at low temperature

$$B_2H_6 + 2NH_3 \longrightarrow B_2H_6.2NH_3$$

$B_2H_6.2NH_3$ on heating to 200^oC gives borazole

$$3B_2H_6.2NH_3 \xrightarrow{200^oC} 2B_3N_3H_6 + 12H_2 \uparrow$$

Moles of addition product = moles of diborane $= \frac{24}{28}$

Moles of borazole $= \frac{2}{3} \times \frac{24}{28}$

Weight of borazole $= \frac{2}{3} \times \frac{24}{28} \times 81 = 46.29$ g

42. C

Sol. $[Ni(CN)_4]^{2-}$ is square planar and coloured.

$[NiCl_4]^{2-}$ is tetrahedral and contains Ni(II) which is coloured.

$K_2[Zn(CN)_4]$ is tetrahedral and colourless.

43. C

Sol. The colloidal solution has positively charged particles.

44. A

Sol. $$NH_4NO_3 \xrightarrow{\Delta} \overset{(+1)}{N_2}O \uparrow + 2H_2O$$

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} \overset{(0)}{N_2} \uparrow + 4H_2O + Cr_2O_3$$

$$Hg(NO_3)_2 \xrightarrow{\Delta} Hg + 2\overset{(+4)}{NO_2} + O_2$$

$$Na(NH_4)HPO_4.4H_2O \xrightarrow{\Delta} NaPO_3 + \overset{(-3)}{NH_3} + 5H_2O$$

+1 + 0 + 4 – 3 = (+ 2)

45. B

Sol. $NaBr(aq) \longrightarrow \underset{(0.01)}{Na^+(aq)} + \underset{(\alpha+0.01)}{Br^-(aq)}$

$AgBr(s) \overset{K_{sp}}{\rightleftharpoons} \underset{\alpha}{Ag^+(aq)} + \underset{(\alpha+0.01)}{Br^-(aq)}$

$K_{sp} = \left[Ag^+\right]\left[Br^-\right]$

$K_{sp} = (\alpha)(\alpha + 0.01) \Rightarrow (\alpha + 0.01 \approx 0.01)$

$\therefore 10^{-10} = \alpha \times 0.01 \Rightarrow \alpha = 10^{-8}$

Solubility in pure water is $\sqrt{K_{sp}} = 10^{-5}$ M

Factor by which solubility changes = $\frac{10^{-8}}{10^{-5}} = 10^{-3}$

$\therefore x = 3$

46. C

Sol. $SO_2 + PCl_5 \longrightarrow \underset{(X)}{SOCl_2} + POCl_3$

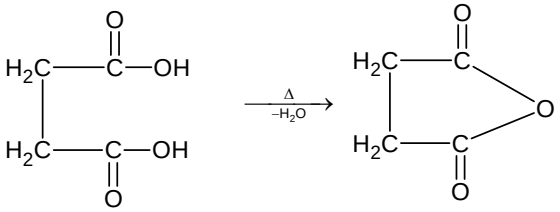
$SOCl_2 + H_2O \longrightarrow SO_2 + \underset{\text{(Pungent smell)}}{2HCl}$

47. C

Sol. Polystyrene, Teflon and Polyacrylonitrile are addition homopolymer, whereas melamine is copolymer.

48. D

Sol. β-Ketoacid, β- γ - unsaturated acid and 1-3 diacid decarboxylates on heating, where as 1-4 diacid dehydrates on heating.



49. D

Sol. Siderite → $FeCO_3$
Anglesite → $PbSO_4$

50. B

Sol. Vitamin C is water soluble, but Vitamin D is fat soluble.
Morphine is used as analgesic.
Vitamin E is known as lactoflavin.

51. A

Sol. MnO_2, is a selective and specific oxidizing agent which oxidizes allylic or benzylic hydroxyl group to carbonyl group (C = O) where as PCC is a mild oxidant which oxidizes 1° and 2° ROH to aldehyde and ketones respectively.

52. C

Sol. Radius (x_1) of nth orbit for H-atom = $r_1 \times n^2$

If n is lower orbit, the next orbit = n + 1

and $x_2 = r_1 (n + 1)^2 = r_1(n^2 + 2n + 1)$

$x_2 - x_1 = r_1 (2n + 1)$

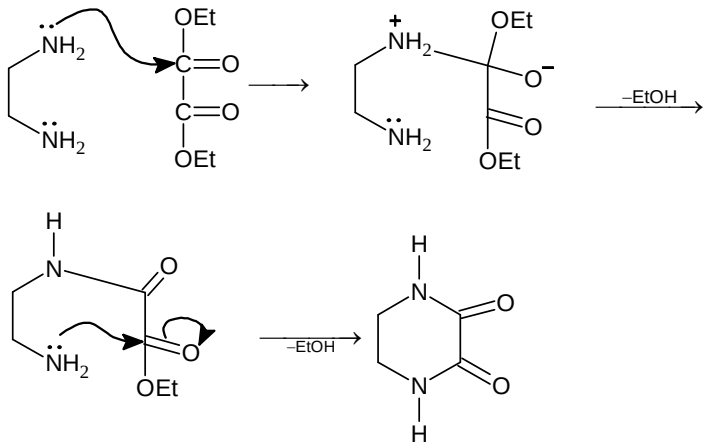
If n is higher orbit, then consecutive lower orbit = n – 1

and $x_2' = r_1(n-1)^2 = r_1(n^2 - 2n + 1)$

$x_1 - x_2' = r_1 \times n^2 - r_1(n^2 - 2n + 1) = r_1 \times (2n - 1)$

53. A

Sol.



54. A

Sol. $Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$

$3Cl_2 + 6NaOH \longrightarrow 5NaCl + NaClO_3 + 3H_2O$

55. B

	A	⟶	B	+	C	+	D
At $t = 0$	400 mm						
After 2 hrs	$400 - x$		x		x		x

Sol. As per question $400 - x + x + x + x = 800$

$\Rightarrow x = 200$

$$k = \frac{2.303}{2}\log\frac{P_A^o}{P_A}$$

$$= \frac{2.303}{2}\log\frac{400}{200}$$

$= 0.346\ hr^{-1}$

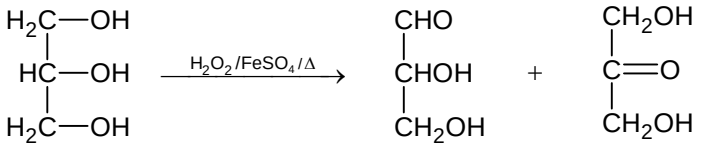
56. D

Sol. D-Glucose and D-Fructose are reducing sugar.

Glycogen and dextrin are non-reducing sugars.

α-D-(+) glucopyranose is an example of hemiacetal.

57. D

Sol. 

Conc. HNO_3 gives mainly glyceric acid.

Dil. HNO_3 oxidizes glycerol to glyceric acid and tartonic acid.

$Bi(NO_3)_3$ gives mainly mesoxalic acid.

58. B

Sol. $\Delta S = nC_V \ell n\frac{T_2}{T_1} + nR\ell n\frac{V_2}{V_1}$

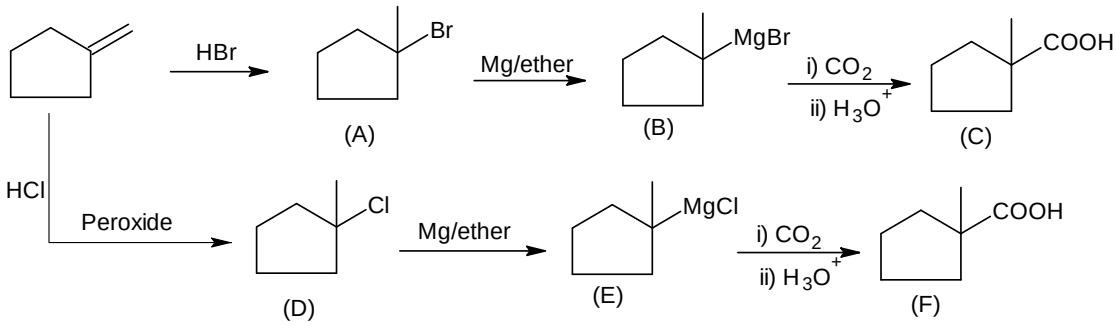
$= 2.303 \times 5 \times 3 \times \log\frac{600}{300}$

$= 10.39$ cal/K

59. B

Sol. German silver, Gun-metal and Bronze are alloys of copper but magnelium is alloy of aluminium and magnesium.

60. B

Sol. 

Mathematics

PART – III

SECTION – A

61. C

Sol. $(9 - x_1) + (9 - x_2) + (9 - x_3) + (9 - x_4) + (9 - x_5) + (9 - x_6) = 49$

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 5$

Number of solution is ${}^{5+6-1}C_{6-1} = {}^{10}C_5$

62. D

Sol. $A^3 = A^2 \cdot A = (2I - A)A = 2A - A^2 = 2A - (2I - A) = 3A - 2I$

$A^4 = A^3 \cdot A = (3A - 2I)A = 3A^2 - 2A = 3(2I - A) - 2A = 6I - 5A$

$A^5 = (6I - 5A)A = 6A - 5A^2 = 6A - 5(2I - A) = 11A - 10I$

$\Rightarrow P = 11$ and $K = -10$

63. B

Sol. $$\int \frac{1-\cos x - x\sin x}{(x-\sin x)^2 + \cos^2 x}\,dx = \int \frac{1}{1+\left(\frac{\cos x}{x-\sin x}\right)^2} \times \frac{1-\cos x - x\sin x}{(x-\sin x)^2}\,dx$$

Let $\frac{\cos x}{x-\sin x} = t \Rightarrow \int \frac{1}{1+t^2}dt = \tan^{-1} t + c = \tan^{-1}\left(\frac{\cos x}{x-\sin x}\right) + c$

64. D

Sol. Around $x = 4 \Rightarrow f(x) = 3 - \sin(x - 2)$

$\therefore$ It continuous and differentiable $f'(x) = -\cos(x - 2)$

65. B

Sol. $$\cot\theta - \frac{\sin\theta}{\cos\theta} = \frac{\cos 2\theta}{\sin\theta\cos\theta} \Rightarrow \frac{\cos 2\theta}{\sin\theta\cos\theta} - \frac{\sin 2\theta}{\cos^2\theta} = \frac{\cos 3\theta}{\cos^2\theta\sin\theta}$$

Similarly, $\frac{\cos 6\theta}{\cos^5\theta\sin\theta} - \frac{\sin 6\theta}{\cos^6\theta} = \frac{\cos 7\theta}{\cos^6\theta\sin\theta}$

66. B

Sol. $\frac{dy}{dx} = y + k$ (where $k = \int_0^1 f(x)dx$)

$y = ce^x - k$ if $f(0) = 1 \Rightarrow c - k = 1$

$k = \int_0^1 f(x)dx \Rightarrow k = \int_0^1 (ce^x - k)dx \Rightarrow 2k = c(e - 1) \Rightarrow k = c\left(\frac{e-1}{2}\right) \Rightarrow c = \frac{2}{3-e}$ and $k = \frac{e-1}{3-e}$

67. D

Sol. $$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2+b^2+c^2}{2abc} = \frac{(a+b+c)^2 - 2(ab+bc+ca)}{2abc} = \frac{144-94}{120} = \frac{5}{12}$$

68. A

Sol. $C_1C_2 = |r_1 + r_2|$ or $C_1C_2 = |r_1 - r_2|$

$10 = |(5 + 0.2t) + (3 + 0.3t)|$ or $10 = |5 + 0.2t - 3 - 0.3t|$

$\Rightarrow t = 4$ sec or $t = 120$ sec

69. B

Sol. $4^{\cot^2 x} = 4$ or $4^{\cot^2 x} = 16$

$\cot^2 x = 1$ or $\cot^2 x = 2$

Number of solution 2 + 2 = 4

70. D

Sol. B is fixed $\ell x + my + n = 0$ passes and we know $\ell - 2m + n = 0$ point B is (1, –2)

$\therefore$ Side length = $\sqrt{52}$

Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times a^2 = \frac{\sqrt{3}}{4} \times 52 = 13\sqrt{3}$

71. C

Sol. Domain $\left[-\frac{1}{2\sqrt{3}}, \frac{1}{3}\right]$ $\therefore$ a = $-\frac{1}{2}$ and b = 1

Range $\left[0, \sqrt{\frac{5\pi}{6}}\right]$ $\therefore$ c = 0 and d = $\frac{1}{\sqrt{6}}$

72. B

Sol. $f_k(\theta) = \sum_{k=0}^{n} \frac{1}{4^k} \sin^2\left(2^k\theta\right)\left(1 - \cos^2\left(2^k\theta\right)\right) = \sum_{k=0}^{n} \frac{1}{4^k}\left[\sin^2\left(2^k\theta\right) - \frac{1}{4}\sin^2\left(2^{k+1}\theta\right)\right]$

$= \sum_{k=0}^{n}\left[\frac{1}{4^k}\sin^2\left(2^k\theta\right) - \frac{1}{4^{k+1}}\sin^2\left(2^{k+1}+1\right)\right] = \sin^2\theta - \frac{1}{4^{n+1}}\sin^2\left(2^{n+1}\theta\right)$

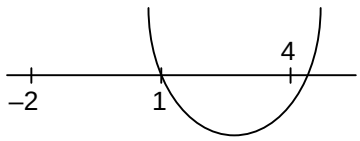
$f_4\left(\frac{3\pi}{2}\right) = \sin^2\frac{3\pi}{2} - \frac{1}{1024}\sin^2\left(32 \times \frac{3\pi}{2}\right) = 1$ $\therefore$ $f_5(\pi) = 0$

73. C

Sol. Let $f(x) = x^2 - 2ax + a^2 - 4$

$\because$ $A \cap B = \{x : -2 < x \le 1\}$

$\therefore$ graph of f(x) must be i.e. $\frac{-b}{2a} > 1$ and $f(4) \le 0$



$\Rightarrow \frac{a}{2} > 1$ and $16 - 8a + a^2 - 4 \le 0$

$\Rightarrow a > 2$ and $a \in a$ is 3 $\therefore$ Least value of a is 3

74. D

Sol. $\frac{(a/b)r^5}{(a/b)r^7} = \frac{1}{4} \Rightarrow r = -2$

75. C

Sol. As $x^2 + 3x + 5 + 0$ has non-real roots so the equations $x^2 + 3x + 5 = 0$ and $3x^2 + px + r = 0$ have both roots common i.e., $\frac{3}{1} = \frac{p}{3} = \frac{r}{5} \Rightarrow p = 9$ and $r = 15$

$\because$ $q = \frac{p+r}{2} \Rightarrow q = 12$

ΔPQR is right angled triangle therefore circumradius is half of length of hypotenuse

$\therefore$ Circumradius = $\frac{15}{2}$ and $\Delta = rs \Rightarrow r = \frac{\Delta}{s} = \frac{54}{18} = 3$

76. C

Sol. $f(n)=\sum_{k=-n}^{-1}\left(\cot^{-1}\frac{1}{k}-\tan^{-1}k\right)+\sum_{k=1}^{n}\left(\cot^{-1}\frac{1}{k}-\tan^{-1}k\right)=\sum_{k=-n}^{-1}\left(\tan^{-1}k+\pi-\tan^{-1}k\right)+0=n\pi$

$\therefore \sum_{n=2}^{10}n\pi+(n-1)\pi=\sum_{n=2}^{10}(2n-1)\pi=99\pi$

77. B

Sol. $\sum_{r=0}^{n}\left(\frac{r+2}{r+1}\right){}^{n}C_r=\sum_{r=0}^{n}\left(\frac{r+1+1}{r+1}\right){}^{n}C_r=\sum_{r=0}^{n}{}^{n}C_r+\sum_{r=0}^{n}\frac{{}^{n}C_r}{r+1}=2^n+\sum_{r=0}^{n}\frac{1}{n+1}\,{}^{n+1}C_{r+1}=2^n+\frac{2^{n+1}}{n+1}$

$=\frac{2^n(n+3)-1}{n+1}\ \therefore n=13$

78. C

Sol. $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)=\prod_{n=3}^{\infty}\left(\frac{n^2-4}{n^2}\right)=\prod_{n=3}^{\infty}\frac{n-2}{n}\times\prod_{n=3}^{\infty}\frac{n+2}{n}$

$\Rightarrow L=\left(\frac{1}{3}\times\frac{2}{4}\times\frac{3}{5}\times.....\times\frac{n-2}{n}\right)\times\left(\frac{5}{3}\times\frac{6}{4}\times\frac{7}{5}\times.....\times\frac{n+1}{n-1}\times\frac{n+2}{n}\right)$

$\Rightarrow L=\frac{1\cdot 2}{n(n-1)}\times\frac{(n+1)(n+2)}{3\times 4};\ \lim_{n\to\infty}\frac{2(n^2+3n+2)}{12(n^2-n)}=\frac{1}{6}$

$M=\prod_{n=2}^{\infty}\frac{n-1}{n+1}\times\frac{n^2+n+1}{n^2-n+1}=\left(\frac{1}{3}\times\frac{2}{4}\times\frac{3}{5}\times\frac{4}{6}\times.....\times\frac{n-2}{n}\times\frac{n-1}{n+1}\right)\times\left(\frac{7}{3}\times\frac{13}{7}\times\frac{21}{3}\times.....\times\frac{n^2-n+1}{n^2+n+1}\right)$

$=n^2+\frac{2}{n(n+1)}\times\frac{n^2\times n+1}{3}=\frac{2}{3}\ \therefore M=\frac{2}{3}$

$N=\prod_{n=1}^{\infty}\frac{(n+1)^2}{n(n+2)}=\prod_{n=1}^{\infty}\frac{n+1}{n+2}\times\prod_{n=1}^{\infty}\frac{n+1}{n}=\frac{2(n+1)}{n+2}=2\ \therefore N=2$

$L^{-1}+M^{-1}+N^{1-}=6+\frac{3}{2}+\frac{1}{2}=8$

79. B

Sol. $(z_1+2)(z_2+2)(z_3+2)=z_1z_2z_3+2(z_1z_2+z_2z_3=z_3z_1)+4(z_1+z_2+z_3)+8$

$=1+2(z_1z_2+z_2z_3+z_3z_1)+4+8=13+2(z_1z_2+z_2z_3+z_3z_1)$

Now $z_1z_2+z_2z_3+z_3z_1=z_1z_2z_3\left(\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right)=\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}$

$=\frac{\bar{z}_1}{|z_1|^2}+\frac{\bar{z}_2}{|z_2|^2}+\frac{\bar{z}_3}{|z_3|^2}=\bar{z}_1+\bar{z}_2+\bar{z}_3=z_1+z_2+z_3=1$

80. B

Sol. Total ways in which 4 customers can order is $5^4=625$

Now for the 4th person, number of ways is ${}^{5}C_1=5$ and for the other three persons, number of ways $=4\times4\times4$

$\therefore$ Total number of ways 4th person orders unordered item $=5\times4^3$

$\therefore$ Probability that 4th person orders an unordered item $=\frac{5\times4^3}{5\times5^4}=\frac{64}{125}$

81. D

Sol. C_1: $(x + 2)^2 + y^2 = 16$

C_2 : $(x – 1)^2 + y^2 = 100$

$CC_1 = r + r_1$ and $CC_2 = r_2 – r$ $\therefore$ $CC_1 + CC_2 = r_1 + R_2 = 14$

$\therefore$ Locus of C is an ellipse whose focus is at C_1 and C_2 and length of major axis = 14

$\therefore$ 2a = 14 and 2ae = 3 $\therefore$ $e = \dfrac{3}{14}$

82. B

Sol. Put x = 0, y = 0, we get f(0) = –4

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{f(x)+f(h)+3xh(x+h)+8xh+4-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(h)+3xh(x+h)+8xh-f(0)}{h} \text{ (as f(0) = –4)}$$

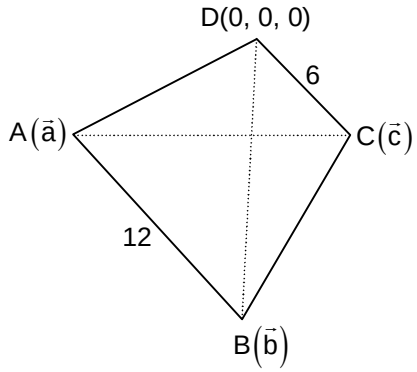
$= f'(0) + 3x^2 + 8x = 3x^2 + 8x$ $\therefore$ $f(x) = x^3 + 4x^2 + c$

as f(1) = 1 $\therefore$ c = –4

$\therefore$ $f(x) = x^3 + 4x^2 – 4$ $\therefore$ $f(x) = x^3 + 4e^x \Rightarrow x^2 – 1 = e^x$ which has one solution

83. C

Sol. Given $|\vec{b} - \vec{a}| = 12$; $|\vec{c}| = 6$



Equation of CD is $\vec{r} = \lambda\vec{c}$

Equation of AB is $\vec{r} = \vec{a} + \mu(\vec{b} - \vec{a})$

Shortest distance between AB and CD

$$= \left|\frac{\vec{a}\cdot(\vec{c}\times(\vec{b}-\vec{a}))}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8 \Rightarrow \left|\frac{[a\ c\ b]}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8$$

$$\left[\vec{a}\ \ \vec{c}\ \ \vec{b}\right] = 8\left|\vec{c}\times(\vec{b}-\vec{a})\right|$$

$\Rightarrow$ 6 $\times$ volume of tetrahedron = $8\times|\vec{c}|\times|\vec{b}-\vec{a}|\times\dfrac{1}{2}$

$\Rightarrow$ v = 48

84. D

Sol. $f'(x) = –((x^2 – x + 1)e^x + e^x(2x – 1)) + e^x \cdot e^{-x} \cdot f'(x) + e^x \int_0^x e^{-y} f'(y)\,dy$

$$f'(x) = f'(x) + \int_0^x e^{x-y} f'(y)\,dy - e^x(x^2 + x)$$

$0 = f(x) + (x^2 – x + 1)e^x – e^x(x^2 + x)$

$f(x) = e^x(x^2 + x – x^2 + x – 1)$

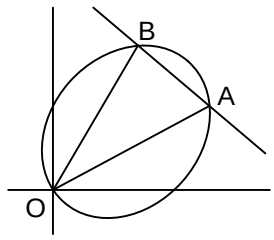
$f(x) = e^x(2x – 1)$

$f'(x) = e^x 2 + (2x – 1)e^x$

$\therefore$ f(1) = e; f′(1) = 3e

85. A

Sol. Equation of circle $x^2 + y^2 + 2gx + 2fy = 0$



Homogenizing the equation of circle with the help of the line $3x + y - 1 = 0$, we get equation of pair of straight lines OA and OB

$x^2 + y^2 + 2gx(3x + y) + 2fy(3x + y) = 0$

$= x^2(1 + 6g) + y^2(1 + 2f) + xy(2g + 6f) = 0$

Comparing it with $23x^2 + 5y^2 + 26xy = 0$

we get $\frac{1+6g}{23} = \frac{1+2f}{5} = \frac{2g+6f}{26}$

$f = -3$ and $g = -4$

$\therefore$ Equation of circle is $x^2 + y^2 - 8x - 6y = 0$

$\therefore$ Radius = 5

86. A

Sol. Any point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$

Equation of the chord of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$

Whose mid-point is $(at^2, 2at)$ is $T = S_1$

$\Rightarrow x\frac{at^2}{2a^2} + y\frac{2at}{a^2} = \frac{a^2t^4}{2a^2} + \frac{4a^2t^2}{a^2} \Rightarrow tx + 4y + at^3 + 8at$

As it passes through $\left(20a, -\frac{a^2}{2}\right) \Rightarrow 20at + 4\left(-\frac{a^2}{2}\right) = at^3 + 8at$

$\Rightarrow at^3 - 12at + 2a^2 = 0 \Rightarrow t^3 - 12t + 2a = 0 \ (a \neq 0)$

Now three should be 3 distinct values of t for three distinct chords

Let $f(t) = t^3 - 12t + 2a$; $f'(t) = 3t^2 - 12 = 0 \Rightarrow t = \pm 2$

So $f(2)\, f(-2) < 0 \Rightarrow (a + 8)(a - 8) < 0$

$\therefore$ 14 values of a are possible $(a \neq 0)$

87. D

Sol. $\int_0^x f(t)dt = x\sqrt{f(x)}$. Differentiating w.r.t. to x, we get $f(x) = \frac{xf'(x)}{2\sqrt{f(x)}} + \sqrt{f(x)}$

$2f(x)^{3/2} = xf'(x) + 2f(x)$; $f'(x) + \frac{2}{x}f(x) = \frac{2}{x}(f(x))^{3/2}$; $\frac{1}{(f(x))^{3/2}}f'(x) + \frac{2}{x}(f(x))^{-1/2} = \frac{2}{x}$

Let $f(x)^{-1/2} = t \Rightarrow \frac{dt}{dx} = -\frac{1}{2}f(x)^{-3/2}f'(x) \Rightarrow -2\frac{dt}{dx} + \frac{2t}{x} = \frac{2}{x} \Rightarrow \frac{dt}{dx}\frac{-1}{x}t = \frac{-1}{x}$

Solving, we get $t = 1 + cx$; i.e., $\frac{1}{\sqrt{f(x)}} = 1 + cx$

$f(1) = \frac{1}{4} \Rightarrow c = 1 \ \therefore \ \sqrt{f(x)} = \frac{1}{1+x} \Rightarrow f(x) = \frac{1}{(1+x)^2}$

Now $f(x) = \text{cosec } x \Rightarrow \frac{1}{(1+x)^2} = \frac{1}{\sin x} \Rightarrow (1 + x)^2 = \sin x$; No solutions

88. B

Sol. Radius of director circle = $\sqrt{a^2 + b^2} = 8 \Rightarrow a^2 + b^2 = 64$

$\therefore 50 + b^2 = 64 \Rightarrow b^2 = 14$; $b^2 = a^2(1 - e^2) \Rightarrow 14 = 50(1 - e^2) \Rightarrow e^2 = \frac{36}{50} \Rightarrow e = \frac{3\sqrt{2}}{5}$

Slope of line joining centre and vertex = $\frac{5-0}{8-3} = -1$ foci lies on this line

$\therefore$ Foci (x, y) can be (x, y) = $\left(8+5\sqrt{2}\times\frac{3\sqrt{2}}{5}\times\frac{-1}{\sqrt{2}},\ 0+5\sqrt{2}\times\frac{3\sqrt{2}}{5}\times\frac{1}{\sqrt{2}}\right) = \left(8-3\sqrt{2},\ 3\sqrt{2}\right)$

or (x, y) = $\left(8-5\sqrt{2}\times\frac{3\sqrt{2}}{5}\times\frac{-1}{\sqrt{2}},\ 0-5\sqrt{2}\times\frac{3\sqrt{2}}{5}\times\frac{1}{\sqrt{2}}\right)$

89. B

Sol. $1 + \text{cosec}^4\,\theta_1 \geq 2$

$2 + \cot^4\,\theta_2 \geq 2$ and $4 + \sin 4\theta_3 \geq 3$

$\therefore$ L.H.S. ≥ 12

$\therefore$ Only possibility is $\sin^2\,\theta_1 = 1$, $\cot^4\,\theta_2 = 0$ and $\sin 4\theta_3 = -1$

$\theta_1 = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$; $\theta_2 = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$; $\theta_3 = \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{11\pi}{8}, \frac{18\pi}{8}$

$\therefore$ Number of triplets are $3 \times 3 \times 4 = 36$

90. A

Sol. $I_1 = \int\limits_{-2017}^{2019} \frac{dx}{\left(5+2x-x^2\right)\left(1+e^{3x-3}\right)}$ ..... (1)

Using the property $\int\limits_a^b f(x)dx = \int\limits_a^b f(a+b-x)dx$, we get

$I_1 = \int\limits_{-2017}^{2019} \frac{dx}{\left(5+2x-x^2\right)\left(1+e^{3-3x}\right)}$ ..... (2)

Adding equation (1) and (2), we get $2I_1 = \int\limits_{-2017}^{2019} \frac{dx}{\left(5+2x-x^2\right)}$

$2I_1 = I_2 \Rightarrow \frac{I_1}{I_2} = \frac{1}{2}$